FLAT BASE CHANGE OF IMMERSIONS: THE COMPARISON MAP OF HIGHER CONORMAL ALGEBRAS IS ONTO AND AN ISOMORPHISM WHEN FLAT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 770-784 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 786-792, the complete author proof (5 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The functoriality map construction for higher conormal algebras (same chapter, lemma-conormal-algebra-functorial) and the schemes-case flat base change result for conormal algebras (Divisors, conormal-algebra-functorial-flat); cited inputs with the etale-localization reduction written in the retained span.

Source reference. Original lemma 0635: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 792–792. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 09RS: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 704–720. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let*

$$\begin{array}{ccc} Z & \xrightarrow{i} & X \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle g} \\ Z' & \xrightarrow{i'} & X' \end{array}$$

be a cartesian square of algebraic spaces over S with i, i' immersions. Then the canonical map $f^*\mathcal{C}_{Z'/X',*} \to \mathcal{C}_{Z/X,*}$ of Lemma 09RS is surjective. If g is flat, then it is an isomorphism.

Proof. We may check the statement after étale localizing X'. In this case we may assume $X' \to X$ is a morphism of schemes, hence Z and Z' are schemes and the result follows from the case of schemes, see Divisors, Lemma 0635. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.